

NUMERICAL METHODS AND OPTIMIZATION

NMO – 3rd Year

ENSIA — Spring 2026

LAB 2

- Numerical Solution of Nonlinear Equations -

Objectives

- Implement the three fundamental numerical methods for solving nonlinear equations.
- Understand the theoretical foundations of nonlinear equation solving.
- Compare convergence speed, numerical stability, and robustness of the methods.
- Estimate experimentally the order of convergence.
- Analyze sensitivity to the initial guess.
- Measure computational performance using CPU time.
- Visualize the geometric interpretation of iterative methods.

Instructions

For each exercise:

- Implement all methods in MATLAB/Octave.
- Use a stopping criterion $\varepsilon = 10^{-8}$ unless otherwise specified.
- Measure CPU time using `tic/toc`.
- Plot error decay in logarithmic scale.

Exercise 1: Classical Nonlinear Equation

Consider:

$$f(x) = x^3 - x - 2$$

Part I – Application of the Methods

1. Plot $f(x)$ on $[1, 2]$.
2. Verify the hypotheses of the Intermediate Value Theorem.
3. Implement:
 - Bisection method
 - Newton–Raphson method
 - Fixed-point iteration using:

$$x = \sqrt[3]{x + 2}$$

4. Derive $f'(x)$ for Newton's method.
5. Test different initial guesses.

Part II – Comparative Study

For each method:

1. Record:
 - Number of iterations
 - Final error
 - CPU time
2. Plot error vs iteration (log scale).
3. Verify:
 - Linear convergence of Bisection
 - Quadratic convergence of Newton
 - Linear convergence of Fixed Point

Part III – Advanced Visualization

1. Animate Newton's method (tangent lines).
2. Plot $g(x)$ and $y = x$ to visualize contraction.
3. Compare geometrical behavior of all methods.

Theoretical Concepts to Verify

- Intermediate Value Theorem
- Banach Fixed-Point Theorem
- Quadratic convergence of Newton
- Linear convergence of Bisection
- Sensitivity to initial guess
- Conditioning in AI-related problems